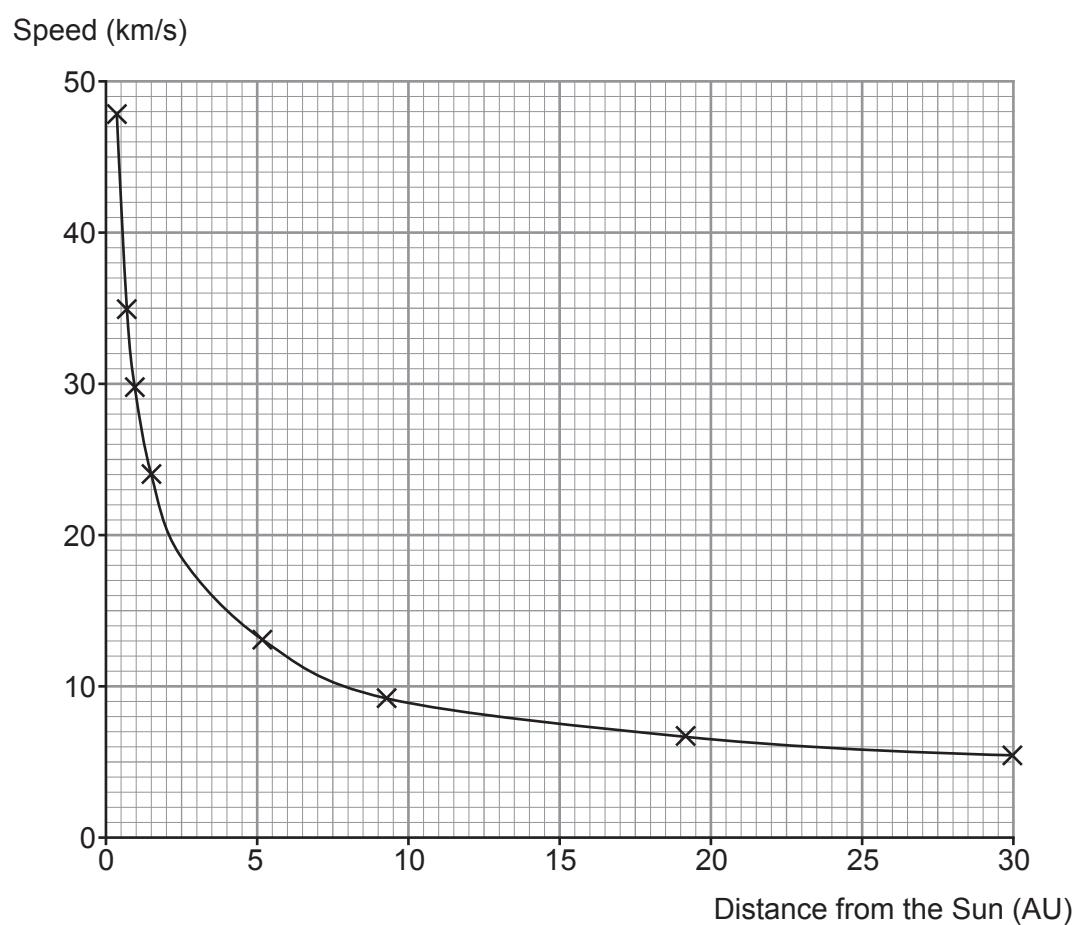


4. There are eight planets in orbit around the Sun in our solar system.
The graph shows how their orbital speeds depend on their distance from the Sun.



- (a) Describe the relationship between the speed of the planets and their distance from the Sun. [2]

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(b) Use the information in the graph to **complete the table**. [2]

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
Speed (km/s)	47.9		29.8	24.1	13	9.7	6.8	5.4
Distance from the Sun (AU)	0.4	0.7	1	1.5	5.2	9.5	19.2	

(c) Name the planet that is about twice as far as Saturn from the Sun. [1]

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(d) The distances on the graph are given in AU (astronomical units).
Use the table to state what distance one AU represents. [1]

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(e) The asteroid belt is between the orbits of Mars and Jupiter.
Paula states that the mean orbital speed of asteroids is 18 km/s.
Owain states that the asteroid belt is 10 AU from the Sun.
Use the data in the table above to explain whether you agree with Paula or with Owain. [3]

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